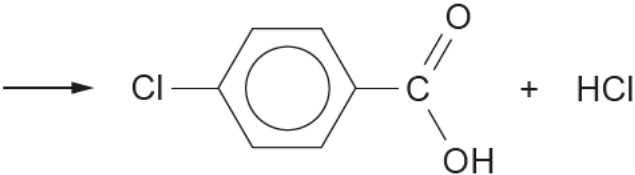
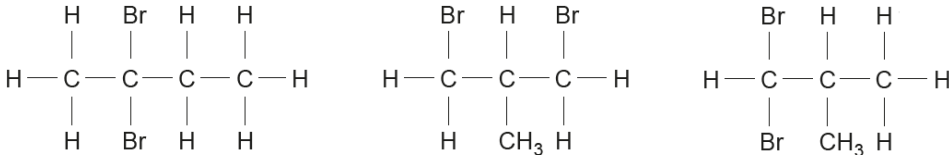
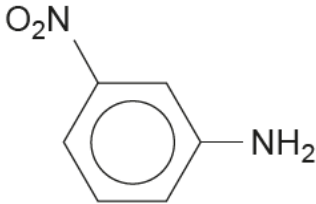
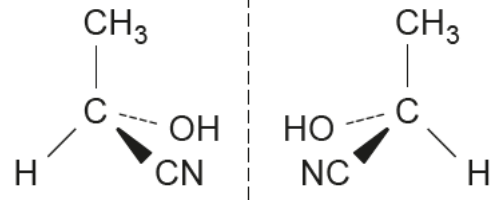


Section A

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
1				C=C at 1620-1670 cm ⁻¹ will identify pent-1-ene		1		1		
2				but-1-ene / but-2-ene / methylpropene correct names are needed here		1		1		1
3						1		1		
4				award (1) for any of following 			1	1		
5						1		1		

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
6	(a)			nucleophilic addition	1			1		
	(b)					1		1		
7	(a)	(i)		$\text{CH}_3\text{---CH}_2\text{---CH}_2\text{---C}\begin{smallmatrix} \text{O} \\ \parallel \\ \text{OH} \end{smallmatrix}$ or $\begin{smallmatrix} \text{H}_3\text{C} \\ \diagup \\ \text{H} \end{smallmatrix}\text{---C---C}\begin{smallmatrix} \text{O} \\ \parallel \\ \text{OH} \end{smallmatrix}$ $\begin{smallmatrix} \text{H}_3\text{C} \\ \diagdown \end{smallmatrix}$	1			1		1
		(ii)		$\begin{smallmatrix} \text{O} & \text{OH} \\ \parallel & \\ \text{CH}_3\text{---C} & \text{---C---CH}_3 \\ & \\ & \text{H} \end{smallmatrix}$			1	1		
	(b)			all the protons are in <u>equivalent environments</u> and the spectrum will consist of <u>one singlet/peak</u>			1	1		
				Section A total	2	5	3	10	0	2